



**ICPC2T
2027**



IEEE

Student Branch
NIT Raipur



IEEE
MADHYA PRADESH SECTION

2027 IEEE International Conference on

Power, Control and Computing Technologies (ICPC2T 2027)

NIT Raipur, Chhattisgarh, India | 3–5 March 2027

SPECIAL SESSION (SS10) ON

Pulsed Power and Plasma Applications: Bridging Electrical Engineering with Healthcare, Food, Agriculture, Defence, Space, and Industry

LIST OF TOPICS

- Fundamentals of PEF and plasma physics in electrical engineering
- Power electronics and control strategies for PEF generation
- Simulation and computational modelling of pulsed power systems
- Healthcare applications: sterilization, wound healing, cancer therapy, drug delivery
- Food processing: microbial inactivation, nutrient preservation, non-thermal pasteurization
- Agriculture: plasma-treated water and seeds, crop yield enhancement, pest control
- Defence: plasma propulsion, electromagnetic shielding, sterilization systems
- Space: plasma propulsion systems, PEF-enabled material processing
- Industry: surface modification, wastewater treatment, eco-friendly manufacturing
- Safety, reliability, and scalability of PEF and plasma systems
- Innovation, policy, and entrepreneurship in pulsed power and plasma technologies

SPECIAL SESSION CHAIRS



Dr. Archana Sharma

ANRF Prime Minister
Professor, NIT Raipur



Prof. Ram Prakash

Department of Physics,
IIT Jodhpur



Prof. R. Sarathi

Dept. of Electrical Engineering,
IITM Chennai

SUBMISSION DETAILS

Submit here

<https://cmt3.research.microsoft.com/User/Login?ReturnUrl=%2FICPCCT2027>

After login, select Special Session (SS10): “Pulsed Power and Plasma Applications: Bridging Electrical Engineering with Healthcare, Food, Agriculture, Defence, Space, and Industry”

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